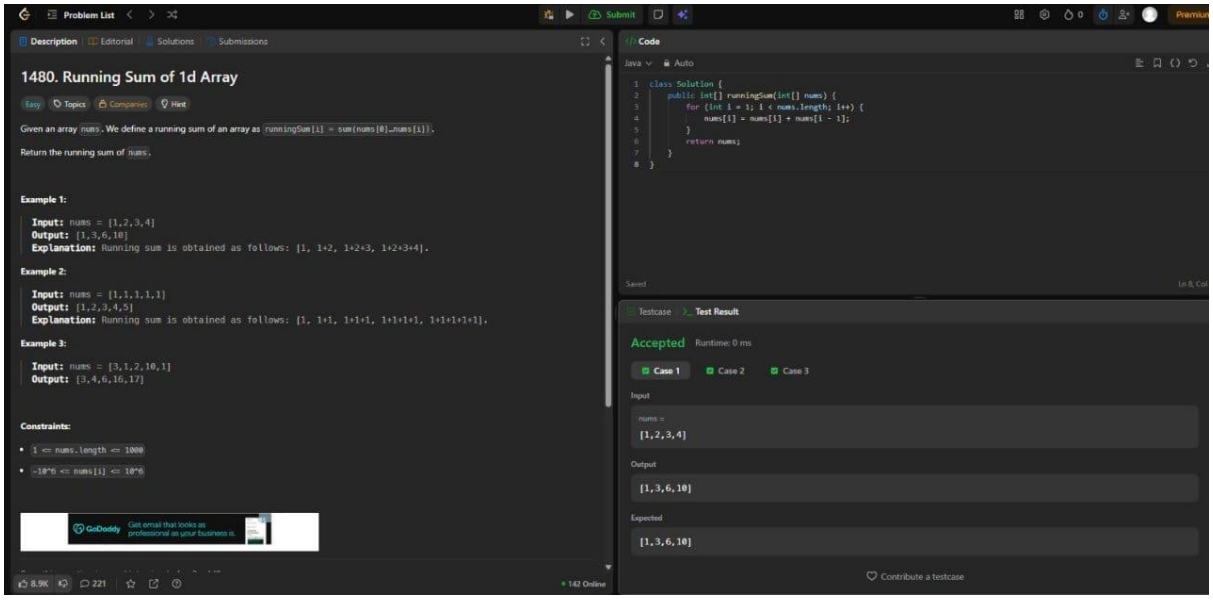
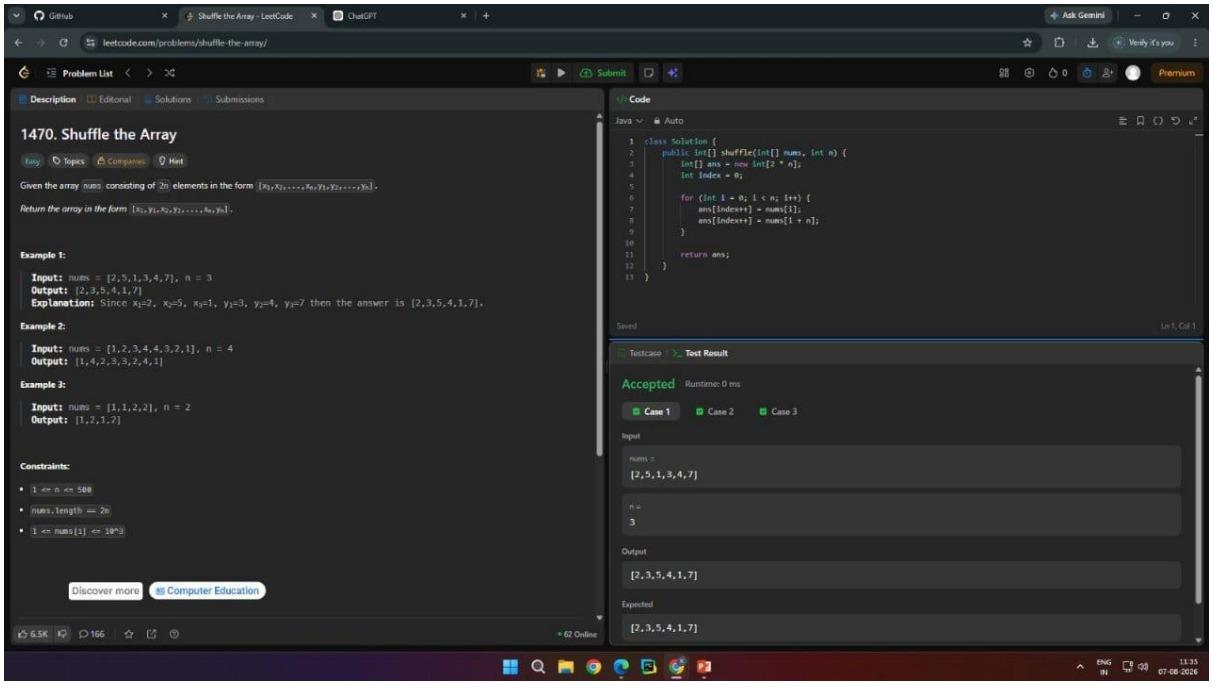
PROBLEM SOLVING AND TESTING USING JAVA

WEEK 2

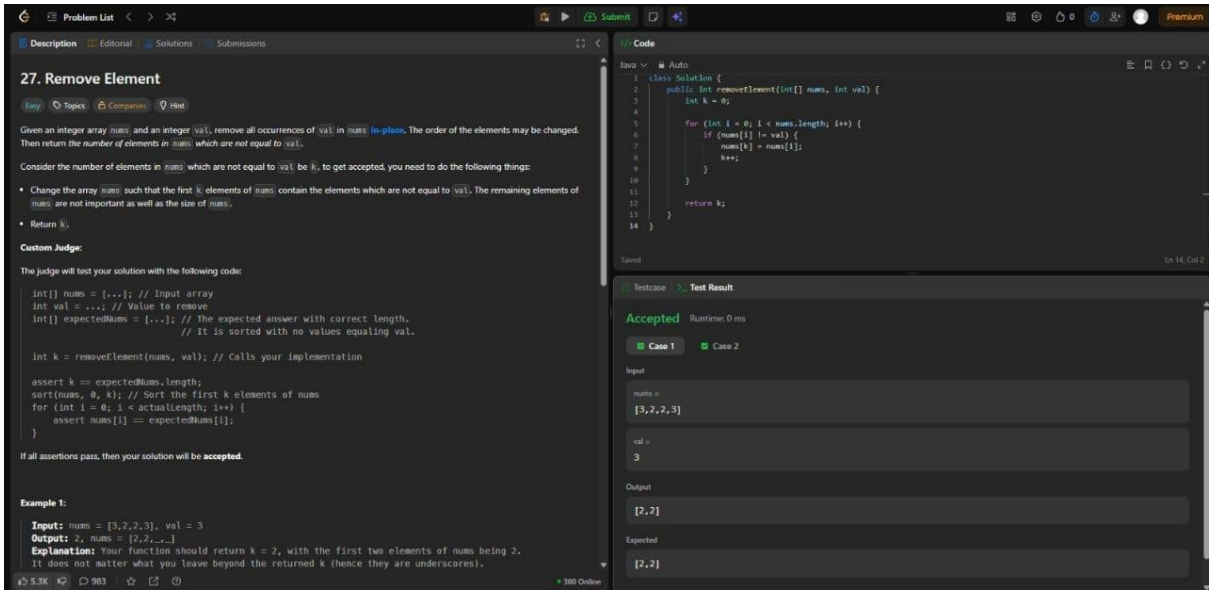
Running Sum of 1D Array



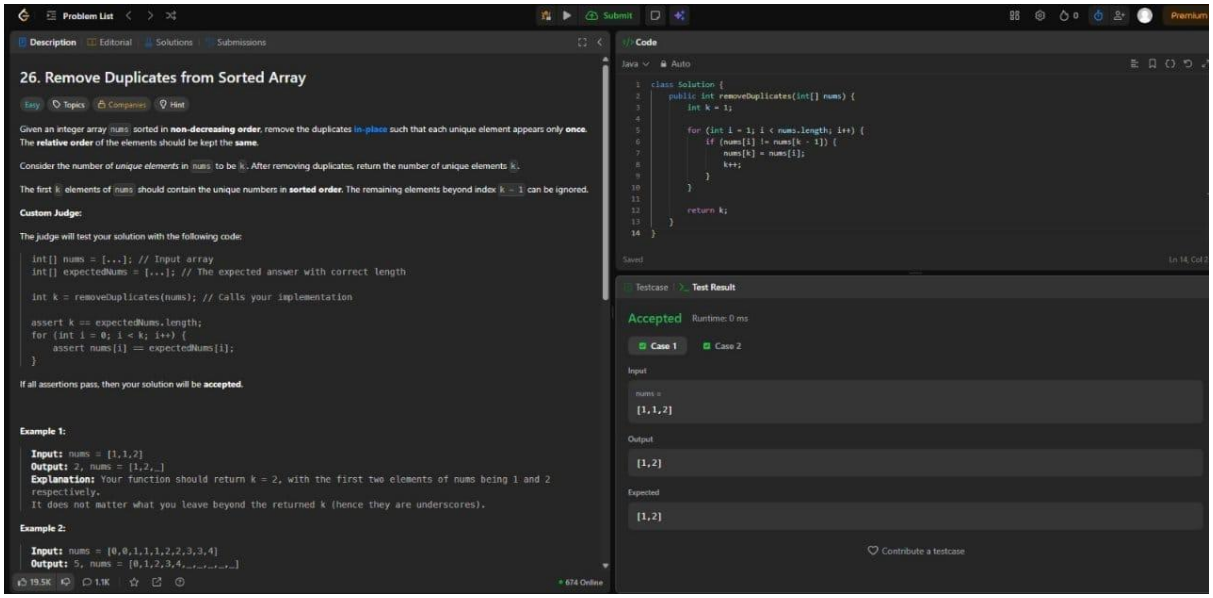
Shuffle The Array



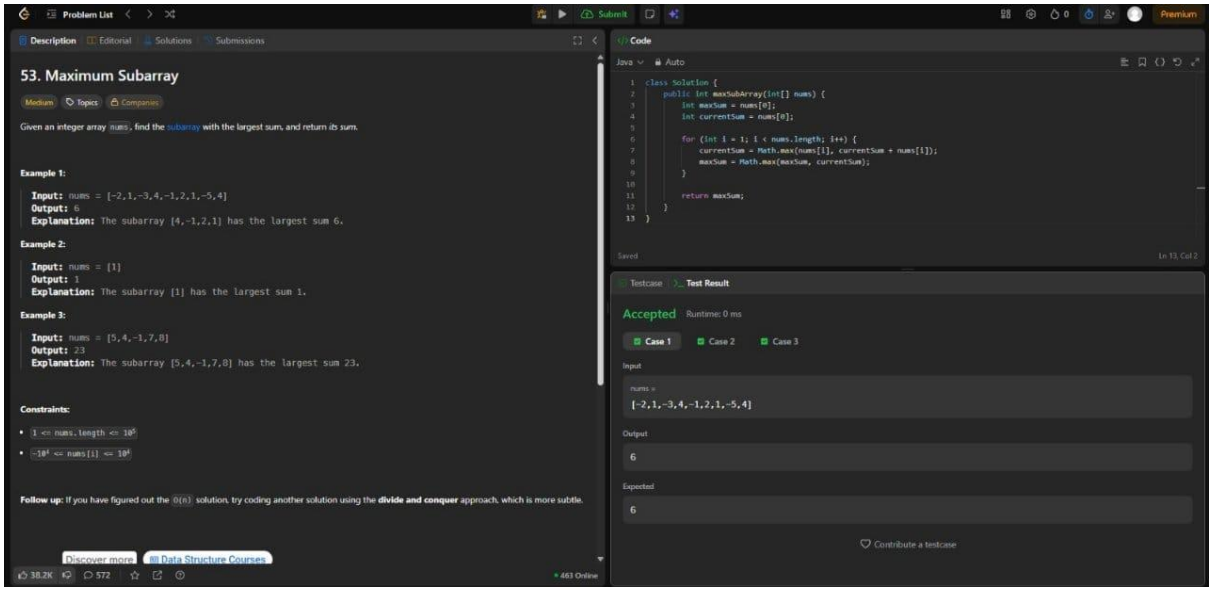
Remove Element



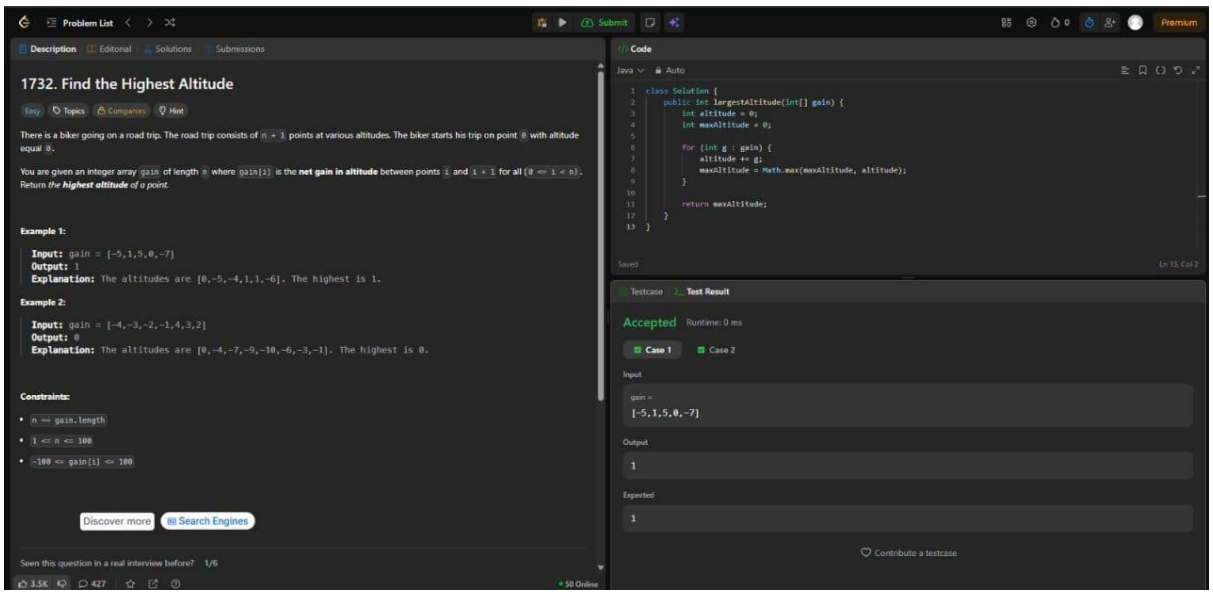
Remove Duplicates From Sorted Array



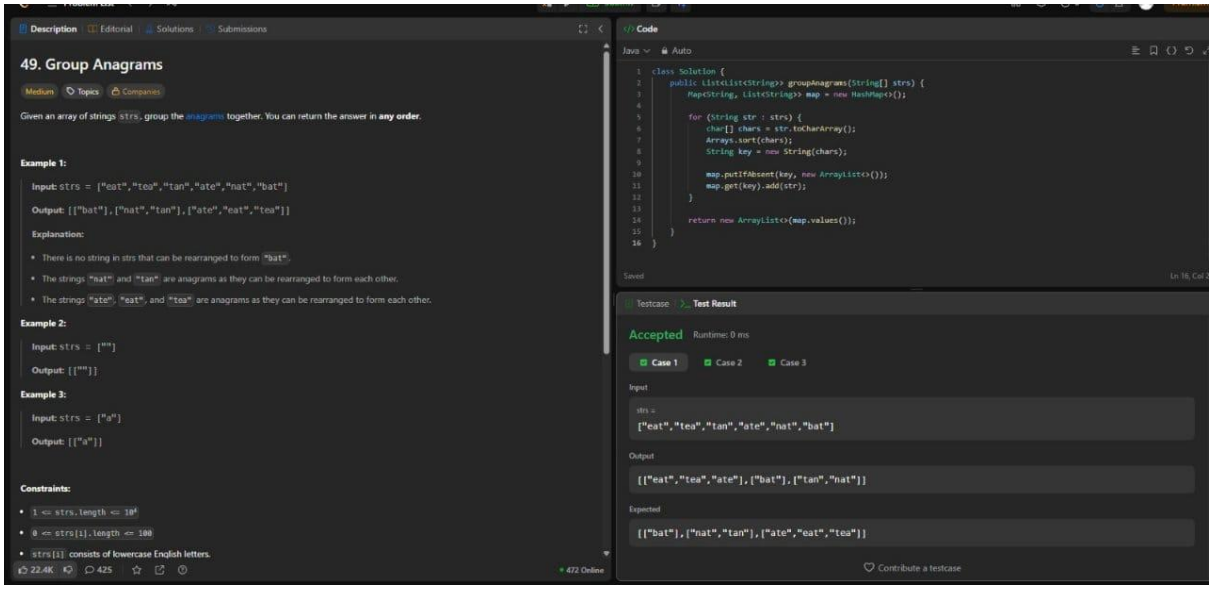
Maximum Subarray



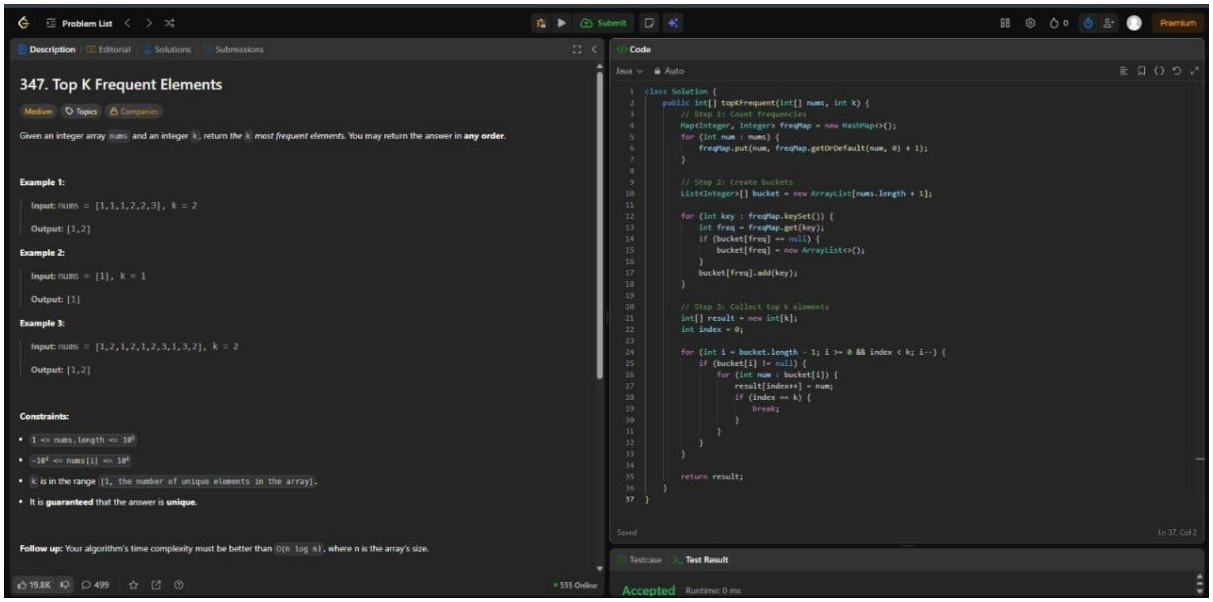
Find The Highest Altitude



Group Anagrams



Top K Frequent Elements



Java Dequeue

```java
import java.util.*;
public class Solution {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int k = sc.nextInt();
        int[] arr = new int[n];
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        HashMap<Integer, Integer> map = new HashMap<>();
        int max = 0;
        for (int i = 0; i < n; i++) {
            if (map.containsKey(arr[i])) {
                map.put(arr[i], map.get(arr[i]) + 1);
            } else {
                map.put(arr[i], 1);
            }
            if (i >= k) {
                int x = arr[i - k];
                if (map.get(x) == 1) {
                    map.remove(x);
                } else {
                    map.put(x, map.get(x) - 1);
                }
            }
            if (i >= k - 1) {
                if (map.size() > max) {
                    max = map.size();
                }
            }
        }
        System.out.println(max);
        sc.close();
    }
}
```

```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

D:\>javac Solution.java

D:\>java Solution
6 3
5 3 5 2 3 2
3

D:\>
```

Java Hashset

```java
import java.util.*;
public class Hashset {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        HashSet<String> set = new HashSet<>();
        for (int i = 0; i < n; i++) {
            String first = sc.next();
            String second = sc.next();
            String pair = first + " " + second;
            set.add(pair);
            System.out.println(set.size());
        }
        sc.close();
    }
}
```

```
C:\Windows\System32\cmd.e
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

D:\>javac Hashset.java

D:\>java Hashset
5
john tom
1
john mary
2
john tom
2
mary anna
3
mary anna
3

D:\>
```